Tim Pengajar IF2150

IF2150 – Rekayasa Perangkat Lunak
Pendahuluan

SEMESTER I TAHUN AJARAN 2025/2026

KNOWLEDGE & SOFTWARE ENGINEERING

Sixty years ago no one could have predicted…

- software would enable the creation of new technologies (e.g., genetic engineering and nanotechnology),
- the extension of existing technologies (e.g., telecommunications),
- the radical change in older technologies (e.g., the media);
- software would be the driving force behind the PC revolution;
- software applications would be purchased by consumers using their smart phones;
- software would slowly evolve from a product to a service as "on-demand" software companies deliver just-in-time functionality via a Web browser;
- a software company would become larger and more influential than all industrial-era companies;
- software-driven network would evolve (from library research to consumer shopping /political discourse / the dating habits).

Software

- Software is designed and built by software engineers.
- Software is used by virtually everyone in society.
- Software is pervasive in our commerce, our culture, and our everyday lives.
- Software engineers have a moral obligation to build **reliable** software that does no harm to other people.
- Software engineers view computer software, as being made up of the **programs**, **documents**, and **data** required to design and build the system.
- Software users are only concerned with whether or not software products meet their **expectations** and make their tasks **easier** to complete.

When computer software succeeds?

- when it **meets the needs of the people** who use it,
- when it **performs flawlessly** over a long period of time,
- when it is **easy to modify** and even **easier to use**

it can and does change things for the better.

When software fails?

- when its **users are dissatisfied**,
- when it is **error prone**,
- when it is **difficult to change** and even **harder to use**

bad things can and do happen

Important Questions for Software Engineers

- Why does it take **so long** to get software finished?
- Why are development **costs so high**?
- Why can't we find all **errors** before we give the software to our customers?
- Why do we spend so much time and effort **maintaining** existing programs?
- Why do we continue to have difficulty in **measuring** progress as software is being developed?

What is software ?

- Definitions:

  Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system
  *(***IEEE Standard Glossary of Software Engineering Terminology, 1990***)*

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Software Characteristics

- Software is both a **product** and a **vehicle** for delivering a product (information).
- Software is **engineered** not manufactured.
- Software does **not wear out**, but it does **deteriorate**.
- Industry is moving toward **component-based software construction**, but most software is still **custom-built**.

Software Application Domains

- **System** software
- **Application** software
- **Engineering** or Scientific Software
- **Embedded** software
- **Product-line** software (includes entertainment software)
- **Web**-Applications
- **Mobile Based** Applications
- **Artificial intelligence** software

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Legacy Software Evolves

- The software must be adapted to meet the needs of new computing environments or technology.
- The software must be enhanced to implement new business requirements.
- The software must be extended to make it interoperable with other more modern systems or databases.
- The software must be re-architected to make it viable within a evolving computing environment.

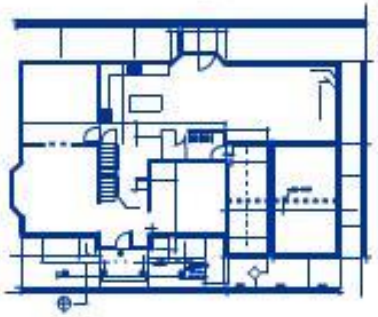
Software Engineering (1)

- Software engineering is the establishment of sound **engineering principles** in order to obtain **reliable** and **efficient** software in an **economical** manner.
- Software engineering is the application of a **systematic**, **disciplined**, **quantifiable** approach to the **development**, **operation**, and **maintenance** of software.
- Software engineering encompasses a **process**, **management techniques**, **technical methods**, and the **use** of tools.

Software Engineering (2)

Engineering:

Design

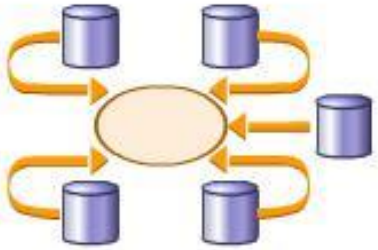


Build

Product

Software Engineering:

Design



Build

Product

Four broad categories of software are evolving to dominate the industry

1. Web-based systems and applications (WebApps)
2. Mobile Applications
3. Cloud computing
4. Product Line Software

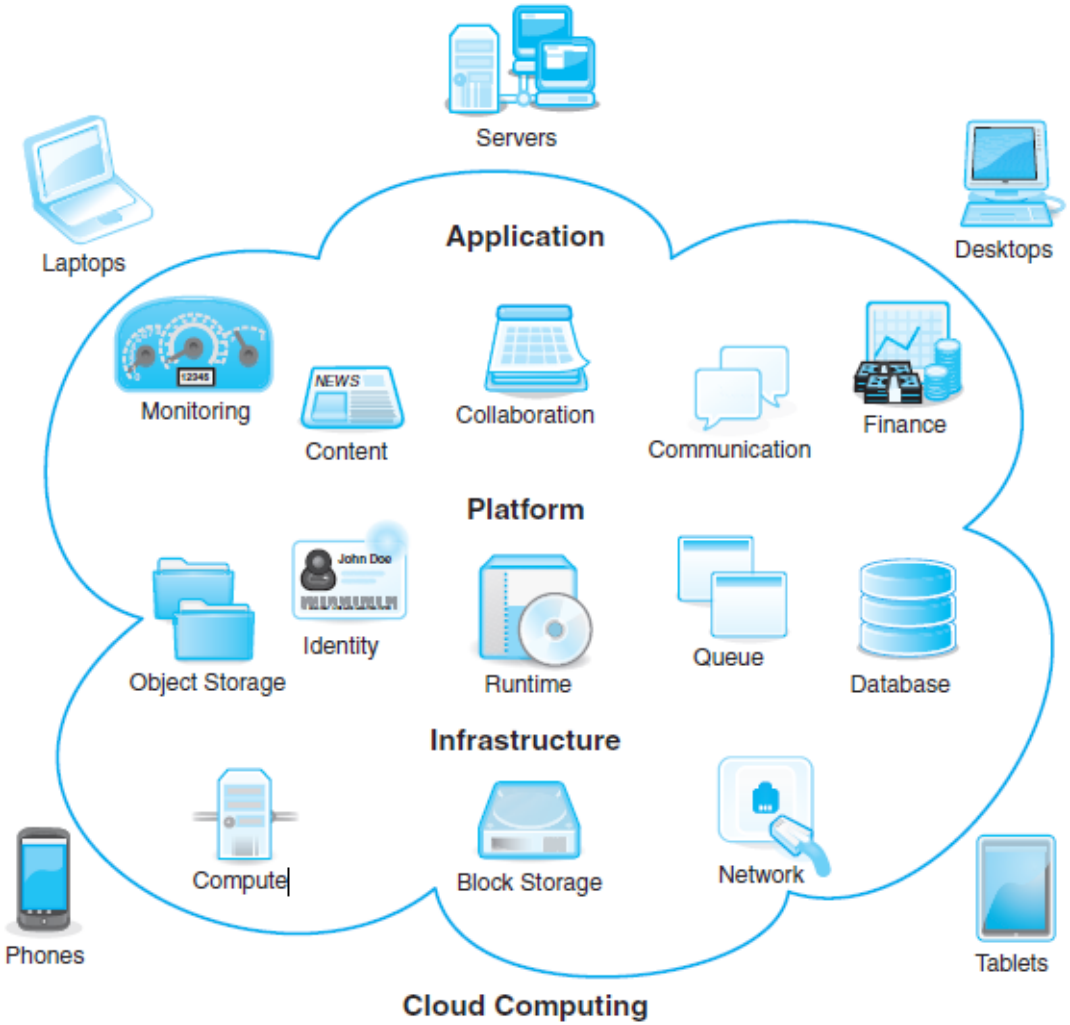
I. Web-based systems and applications

- The augmentation of HTML by development tools (e.g., XML, Java) enabled Web engineers to provide computing capability along with informational content.
- Over the past decade, Semantic Web technologies (Web 3.0) have evolved into sophisticated corporate and consumer applications that encompass "semantic databases [that] provide new functionality that requires Web linking, flexible [data] representation, and external access APIs."
- Sophisticated relational data structures will lead to entirely new WebApps that allow access to disparate information in ways never before possible.

2. Mobile Applications

- The term *app has evolved to connote software that has been specifically designed* to reside on a mobile platform (e.g., iOS, Android, or Windows Mobile).
- encompass a user interface that takes advantage of the unique interaction mechanisms provided by the mobile platform,
- interoperability with Web-based resources

3. Cloud computing



4. Product Line Software

- The Software Engineering Institute defines a *software product line as* "*a set of* software-intensive systems that share a common, managed set of features satisfying the specific needs of a particular market segment or mission and that are developed from a common set of core assets in a prescribed way."
- include requirements, architecture, design patterns, reusable components, test cases, and other software engineering work products

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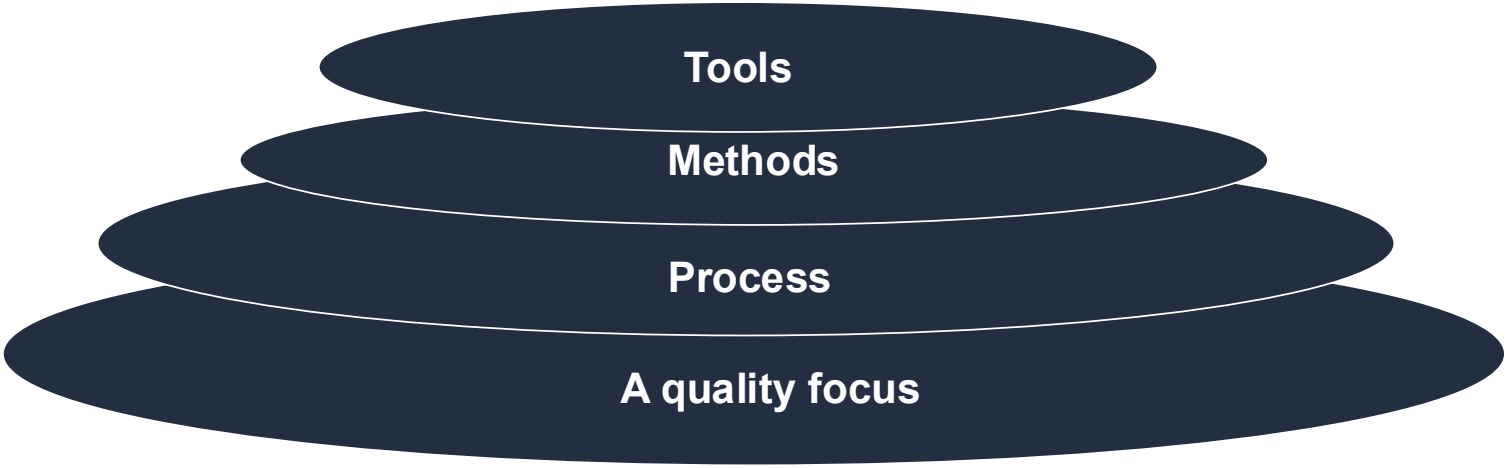
Software Engineering – a layered technology



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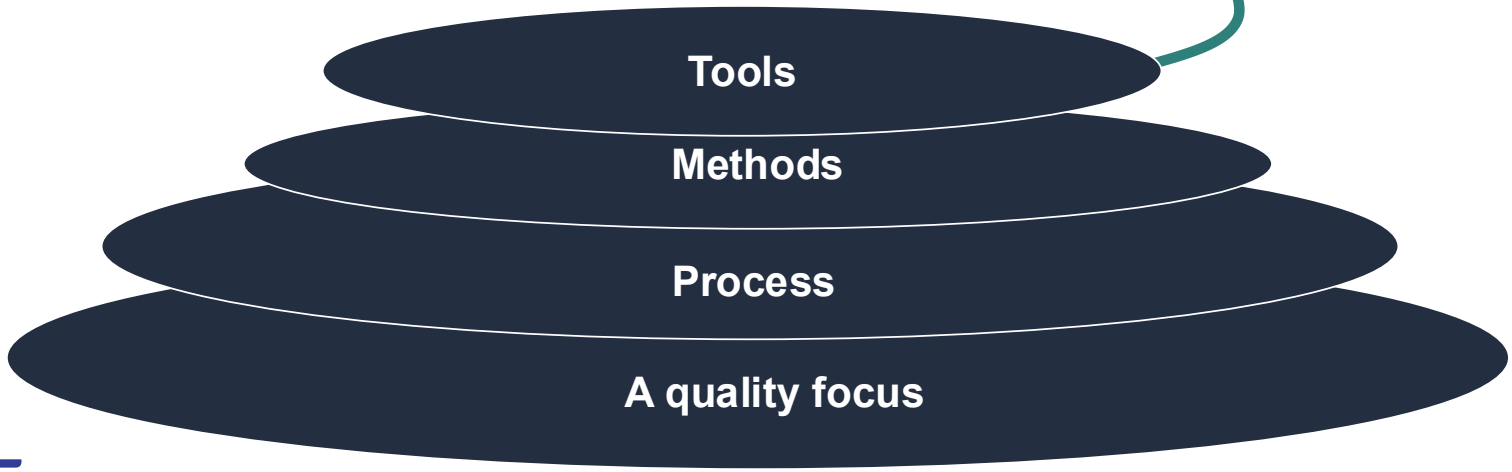
Software Engineering – a layered technology (2)

- The foundation for software engineering is the ***process*** *layer,* defines a framework that must be established for effective delivery of software engineering technology.
- Software engineering ***methods*** *provide the technical how-to's for building* software.
- Software engineering ***tools*** *provide automated or semi-automated support* for the process and the methods
  - *computer-aided software engineering* : e.g Rational Rose; various IDE (Integrated Development Environment) such as: VisualStudio, Eclipse, NetBeans; Software version, such as: CVS, SVN, and GitHub

Software Engineering – a layered technology (3)

- CASE Tool: Rational Rose,
- IDE (Integrated Development Environment): VisualStudio, Eclipse, NetBeans
- Versioning Software: CVS, SVN, GitHub,



CASE tools (Computer-Aided Software Engineering)

- Software systems that are intended to provide automated support for software process activities
- CASE systems are often used for method support
- Upper-CASE
  - Tools to support the **early process** activities of **requirements** and **design**
- Lower-CASE
  - Tools to support **later activities** such as **programming**, **debugging** and **testing**

** Software Engineering 7th ed, Ian Sommerville*

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Case Tools – example

Java - CursorableLinkedList.java - Eclipse SDK

File Edit Source Refactor Navigate Search Project Run Window Help

JUnit

Finished after 34,898 seconds

Runs: 13009/13009 Errors: 0 Failures: 0

Failures | Hierarchy

junit.framework.TestSuite
- junit.framework.TestSuite
  - TestBagUtils
  - org.apache.commons.collections.TestClos
  - org.apache.commons.collections.TestColle
  - TestBufferUtils
  - TestEnumerationUtils
  - org.apache.commons.collections.TestFact
  - TestListUtils
  - TestMapUtils
  - org.apache.commons.collections.TestPrec
  - TestSetUtils
  - org.apache.commons.collections.TestTrar
  - TestArrayStack
  - TestBeanMap
  - org.apache.commons.collections.TestBina
  - TestBoundedFifoBuffer
  - TestBoundedFifoBuffer2
  - TestCursorableLinkedList
  - TestDoubleOrderedMap
  - org.apache.commons.collections.TestExte
  - TestFastArrayList
  - TestFastArrayList1
  - TestFastHashMap
  - TestFastHashMap1
  - TestFastTreeMap
  - TestFastTreeMap1

Failure Trace

CursorableLinkedList.java

```
public boolean addAll(int index, Collection c) {
    if(c.isEmpty()) {
        return false;
    } else if( size == index ||  size == 0) {
        return addAll(c);
    } else {
        Listable succ = getListableAt(index);
        Listable pred = (null == succ) ? null : succ.prev();
        Iterator it = c.iterator();
        while(it.hasNext()) {
            pred = insertListable(pred,succ,it.next());
        }
        return true;
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

TestAllPackages (31.10.2006 15:04:14)

Element	Coverage	Covered Lines	Total Lines
java - commons-collections	79,5 %	10927	13738
org.apache.commons.collections	74,1 %	3842	5183
ArrayStack.java	86,5 %	32	37
BagUtils.java	86,7 %	13	15
BeanMap.java	72,4 %	155	214
BinaryHeap.java	87,6 %	127	145
BoundedFifoBuffer.java	93,2 %	82	88
BufferOverflowException.java	55,6 %	5	9
BufferUnderflowException.java	88,9 %	8	9
BufferUtils.java	30,8 %	4	13
ClosureUtils.java	93,9 %	31	33
CollectionUtils.java	92,4 %	293	317
ComparatorUtils.java	8,6 %	3	35
CursorableLinkedList.java	85,4 %	444	520

Writable | Smart Insert | 149 : 28

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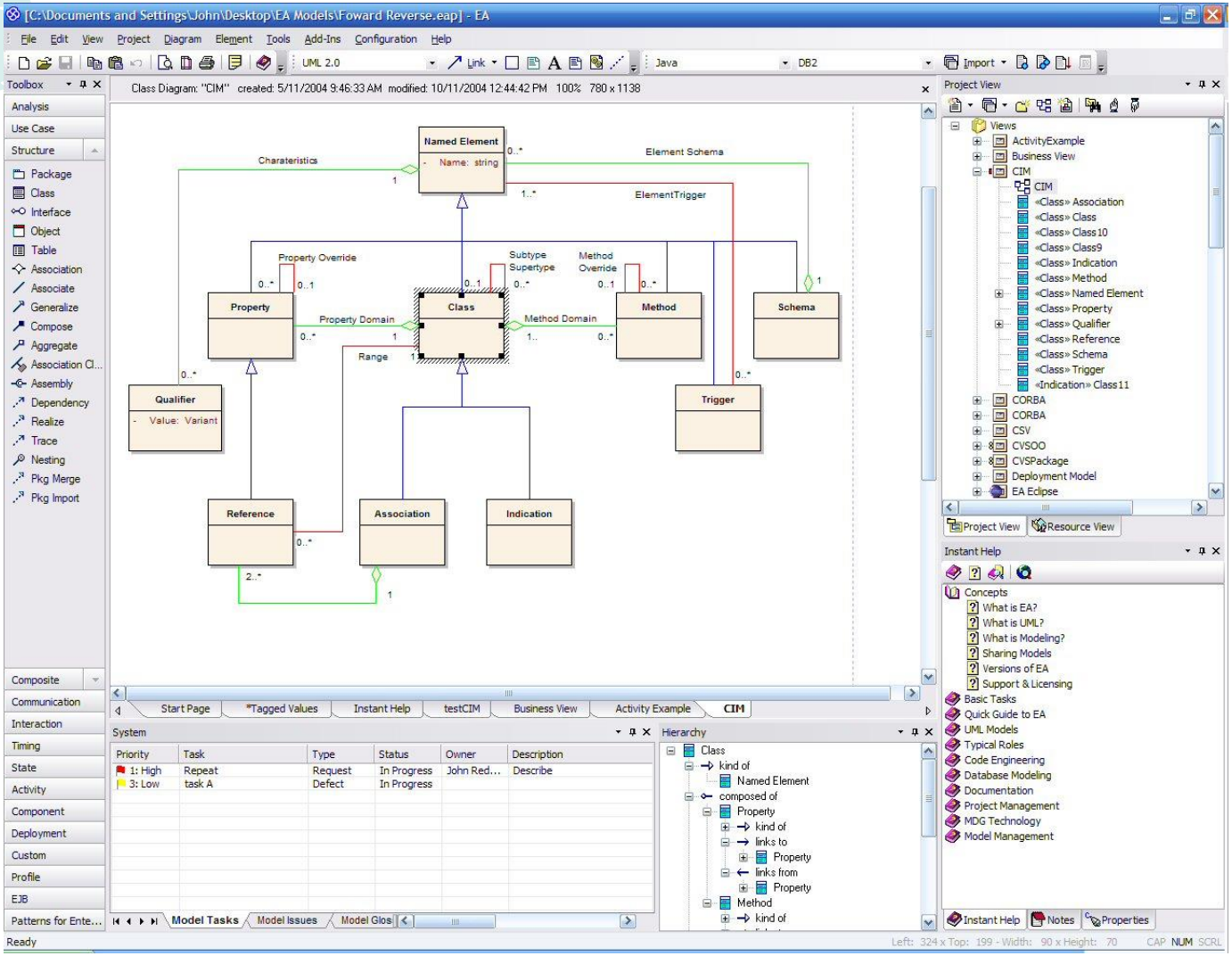
Case Tools – example



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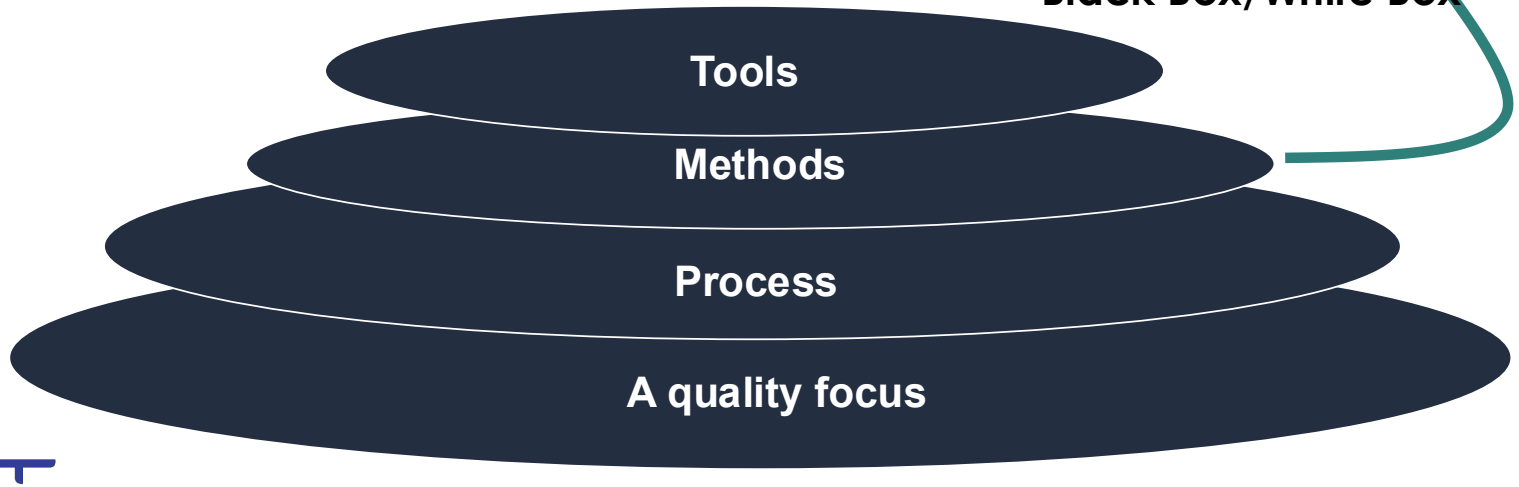
Software Engineering – a layered technology (4)

- Requirement gathering methods
  - Goal Oriented, Viewpoints, etc
- Analysis methods
  - Structured/OO
- Design methods
  - Structured/OO
- Testing methods
  - Black Box/White Box



What are software engineering methods?

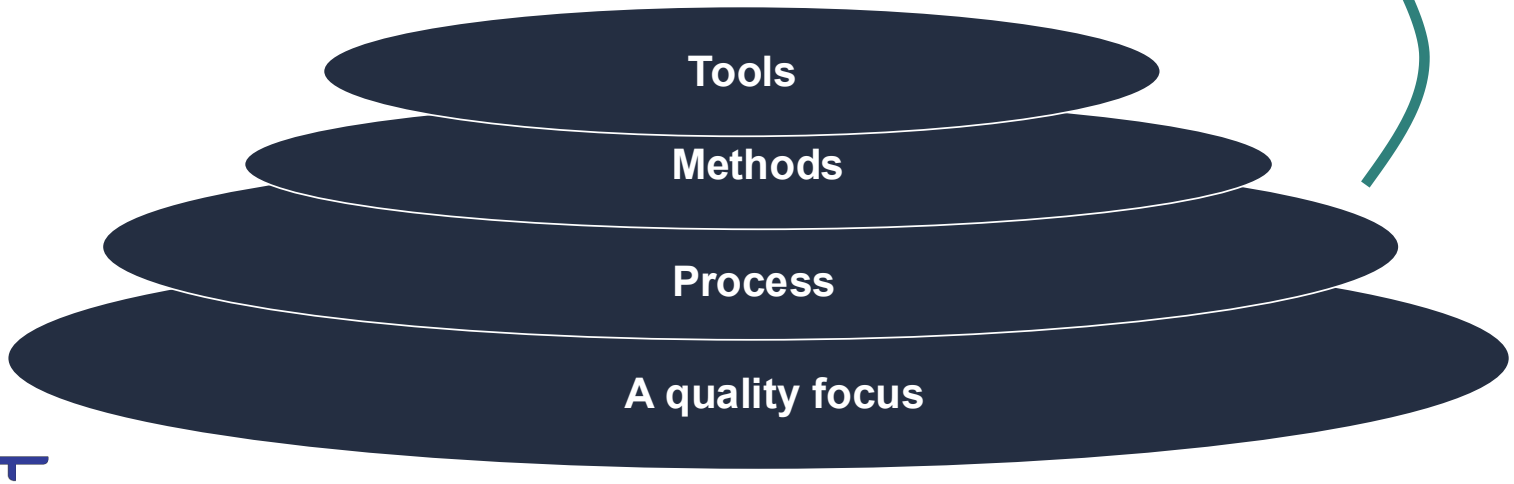
- Structured approaches to software development which include system models, notations, rules, design advice and process guidance.

- Model descriptions
  - Descriptions of graphical models which should be produced
- Rules
  - Constraints applied to system models
- Recommendations
  - Advice on good design practice
- Process guidance
  - What activities to follow

** Software Engineering 7th ed, Ian Sommerville*

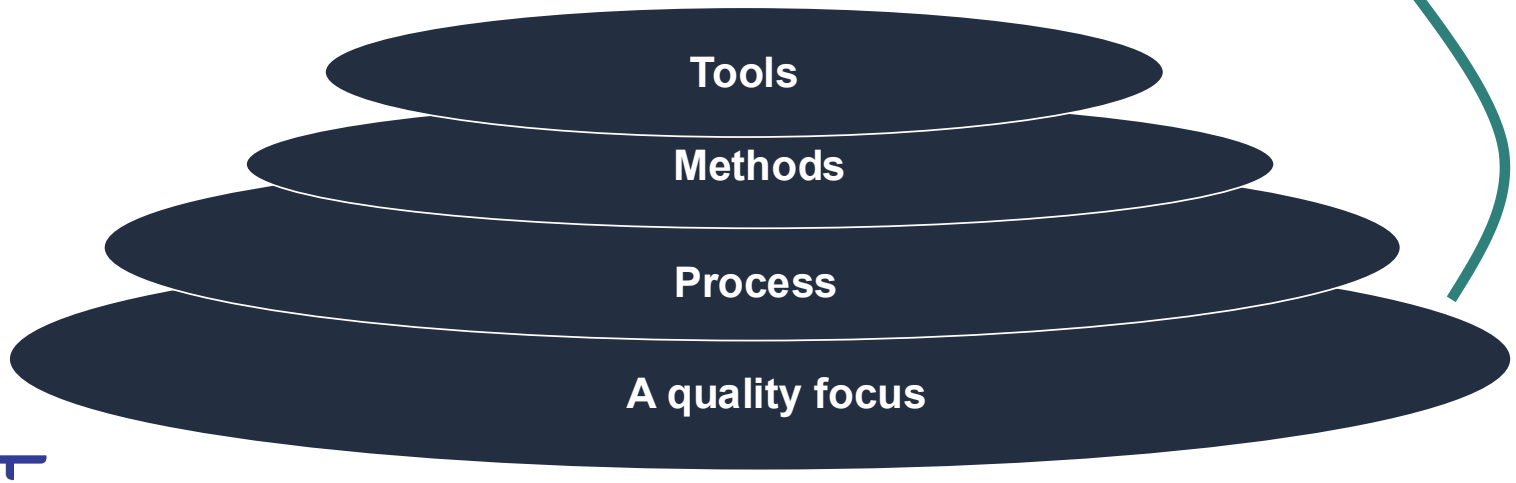
Software Engineering – a layered technology (5)

- Waterfall Model
- Incremental Model/Incremental Process
- Spiral model
- Agile Development
- Rapid Application Development



Software Engineering – a layered technology (6)

- Six Sigma
- Total Quality Management
- CMM (Capability Maturity Model)
- ISO/IEC 9126



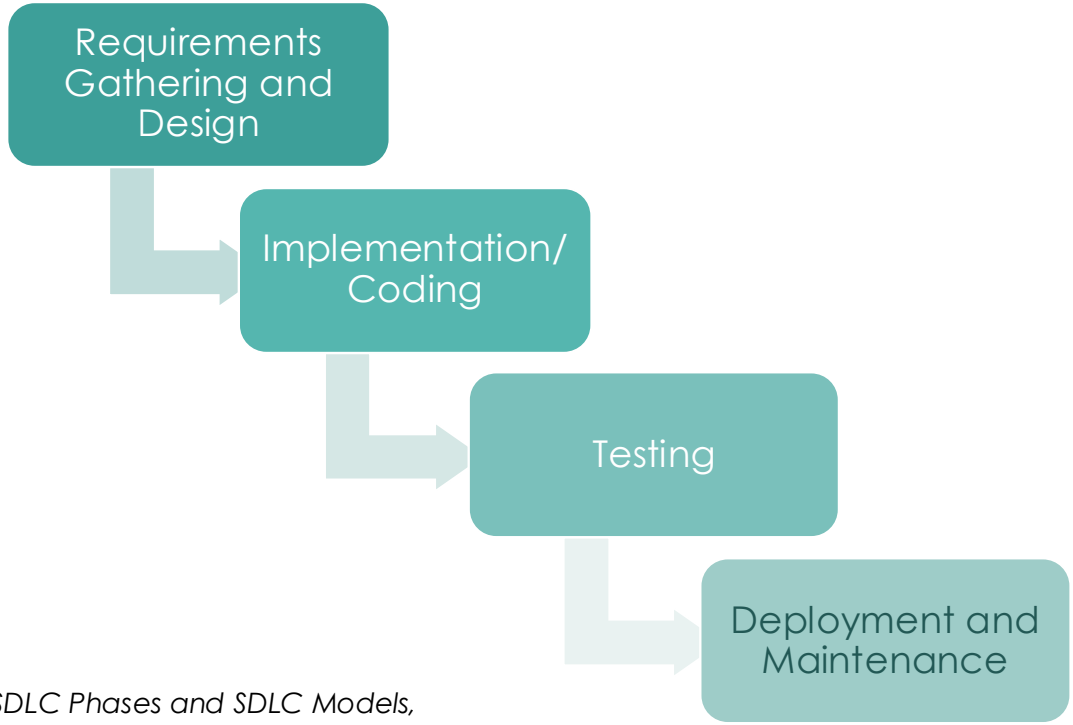
Software Development Activities

Typical Software Development Life Cycle (SDLC) Phases

While specific phases vary by model, a general SDLC often includes:

- **Planning:** Initial project planning and feasibility studies.
- **Requirements Gathering and Analysis:** Defining and analyzing user needs.
- **Design:** Creating the overall system architecture and detailed software design.
- **Development/Implementation:** Writing the actual code for the software.
- **Testing:** Verifying and validating the software to identify and fix defects.
- **Deployment:** Releasing the software to the production environment for users.
- **Maintenance:** Ongoing support, bug fixes, and enhancements after deployment.

A very basic SDLC



BrowserStack: Introduction to SDLC Phases and SDLC Models,
By Vivek Mannotra, Community Contributor - November 26, 2024

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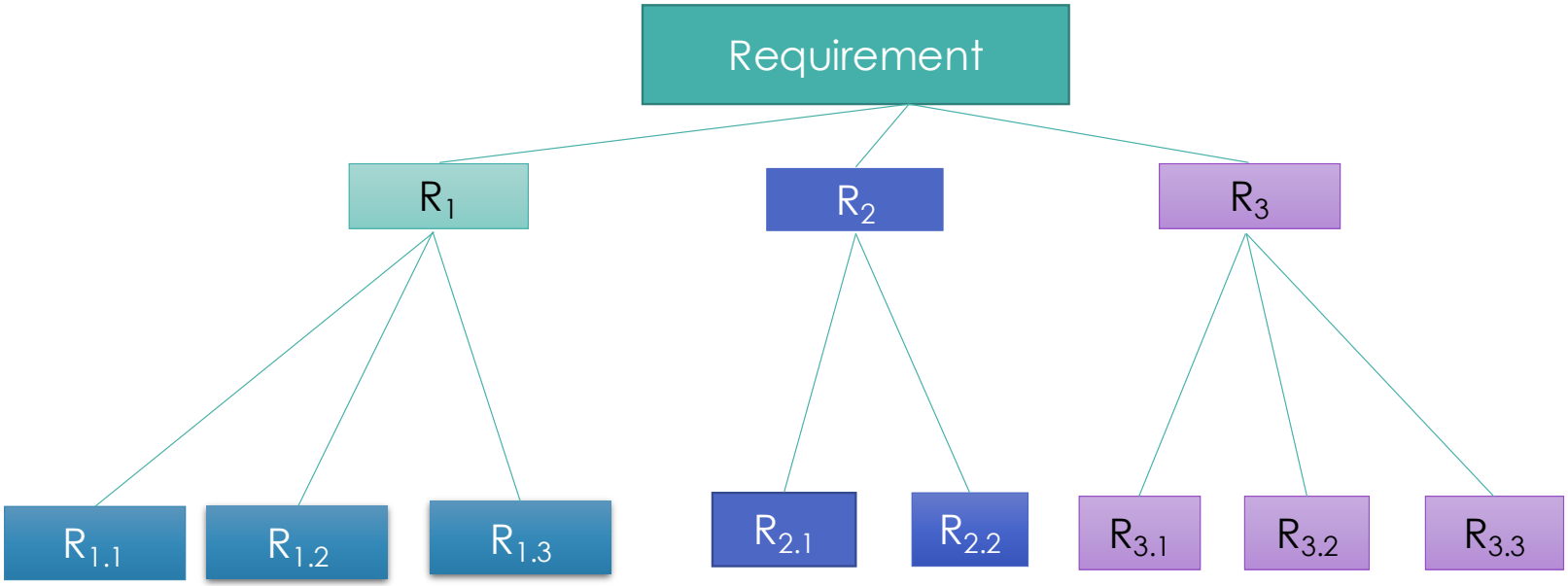
Requirements Gathering



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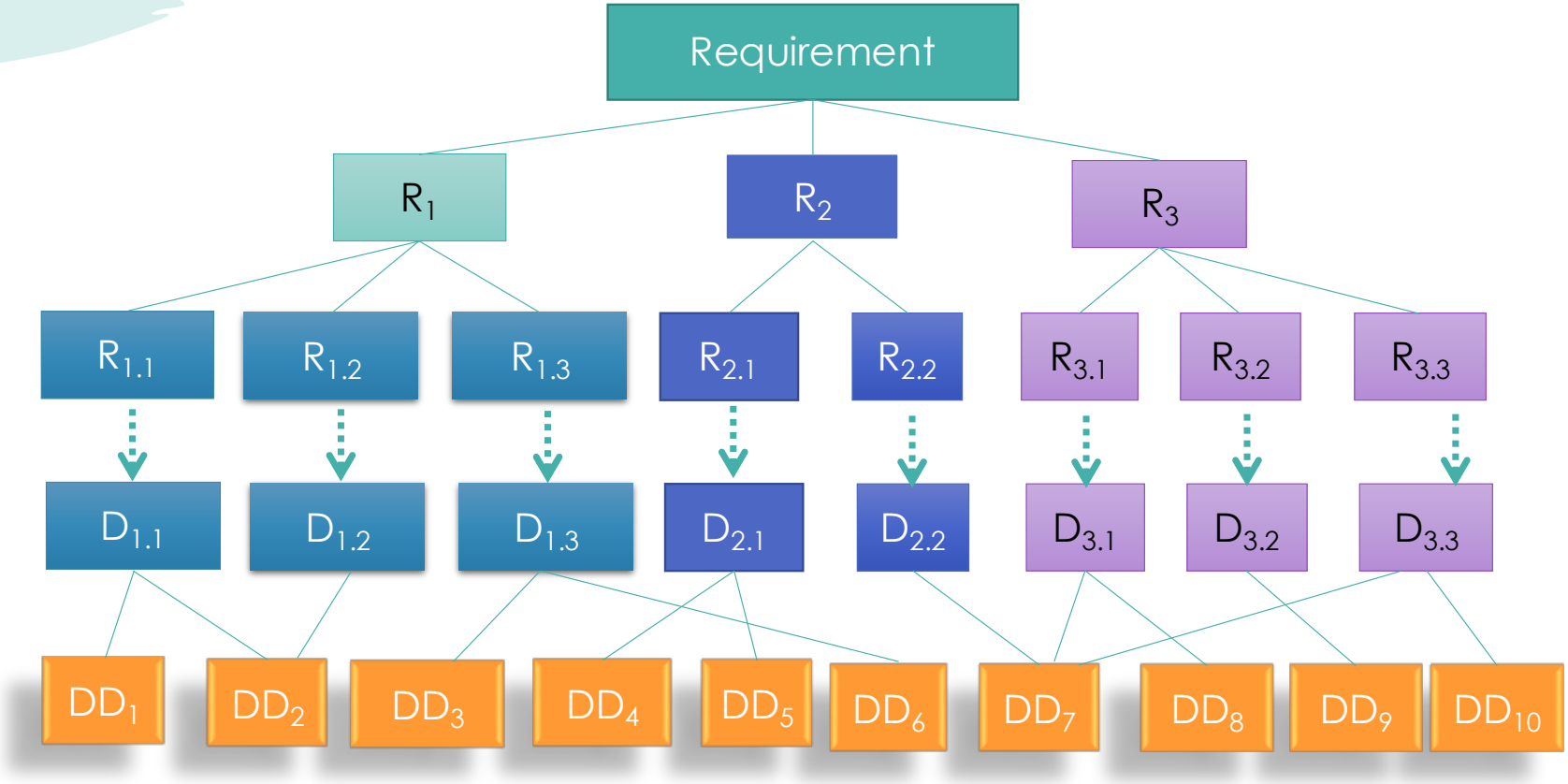
Designing the requirement…



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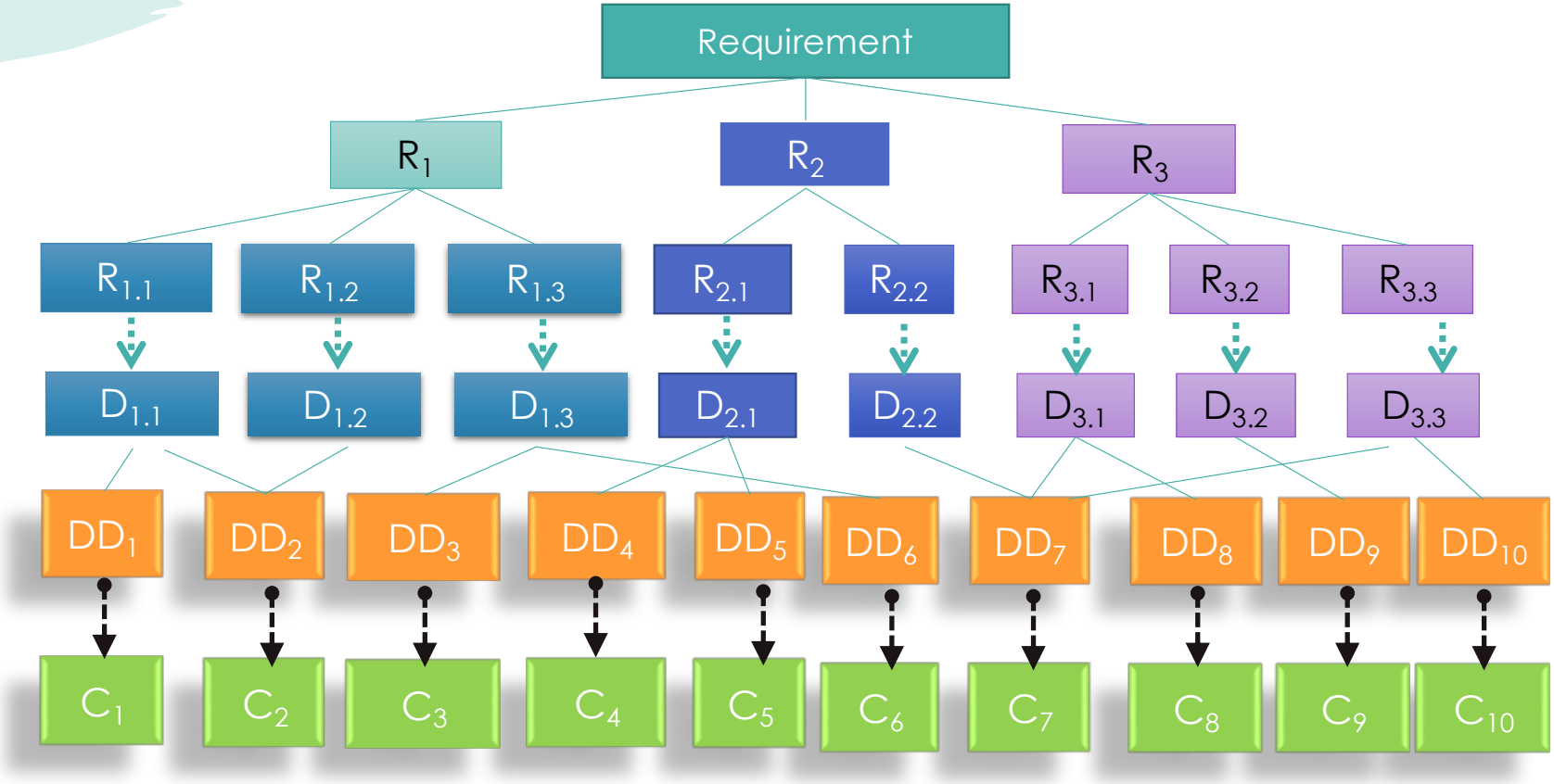
From Design to Coding…



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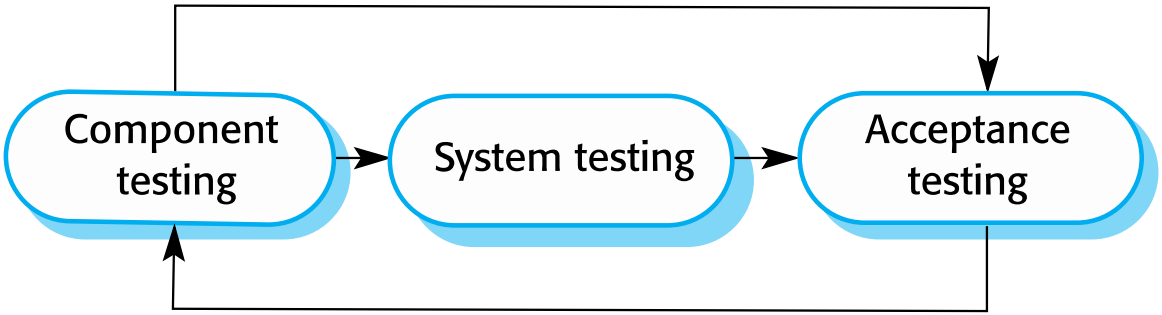
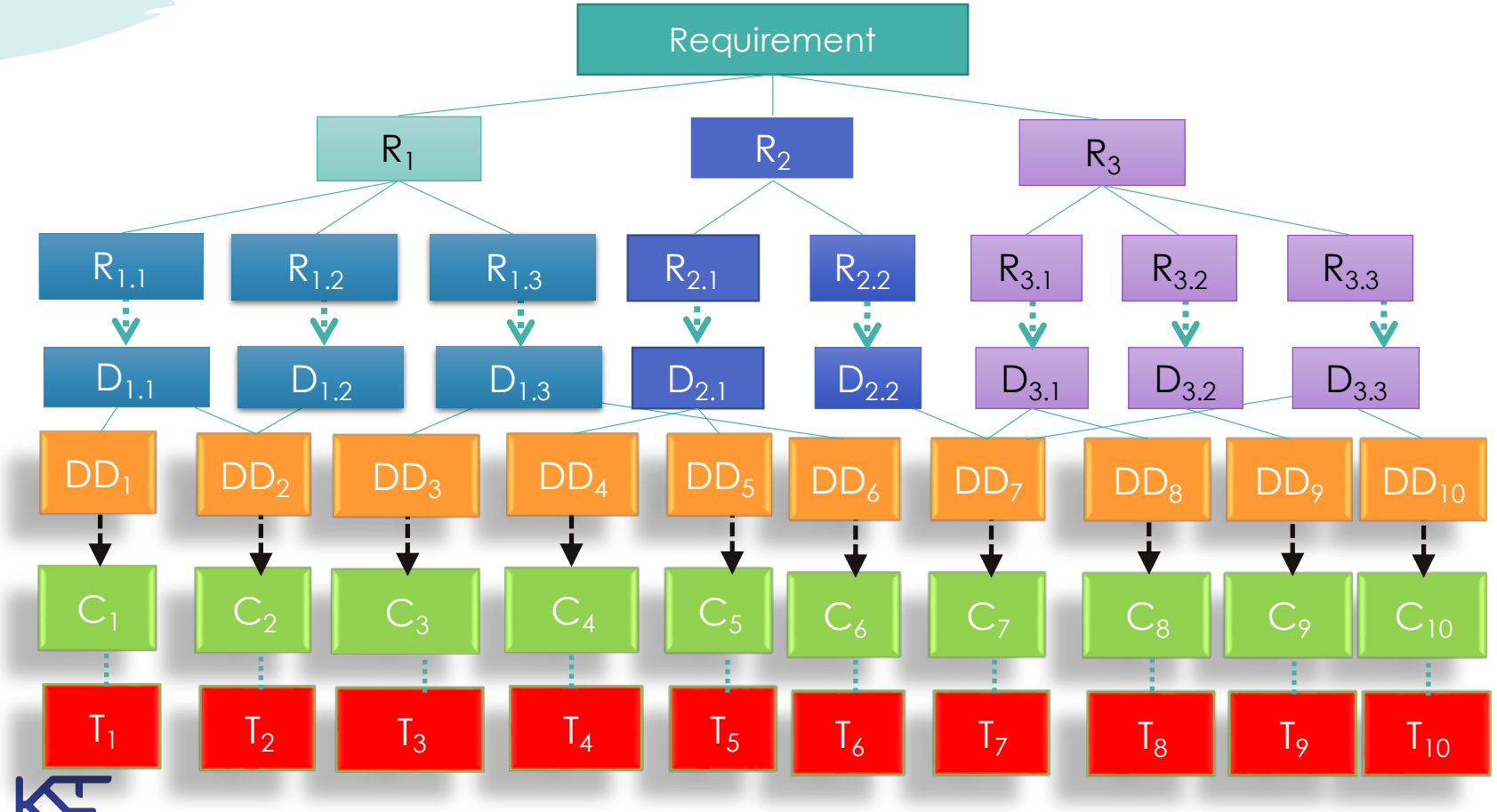
Stages of testing



Testing stages

- Component testing (Unit testing)
  - Individual components are tested independently;
  - Components may be functions or objects or coherent groupings of these entities.
- System testing
  - Testing of the system as a whole. Testing of emergent properties is particularly important.
- Customer testing
  - Testing with customer data to check that the system meets the customer's needs.

Unit Testing…



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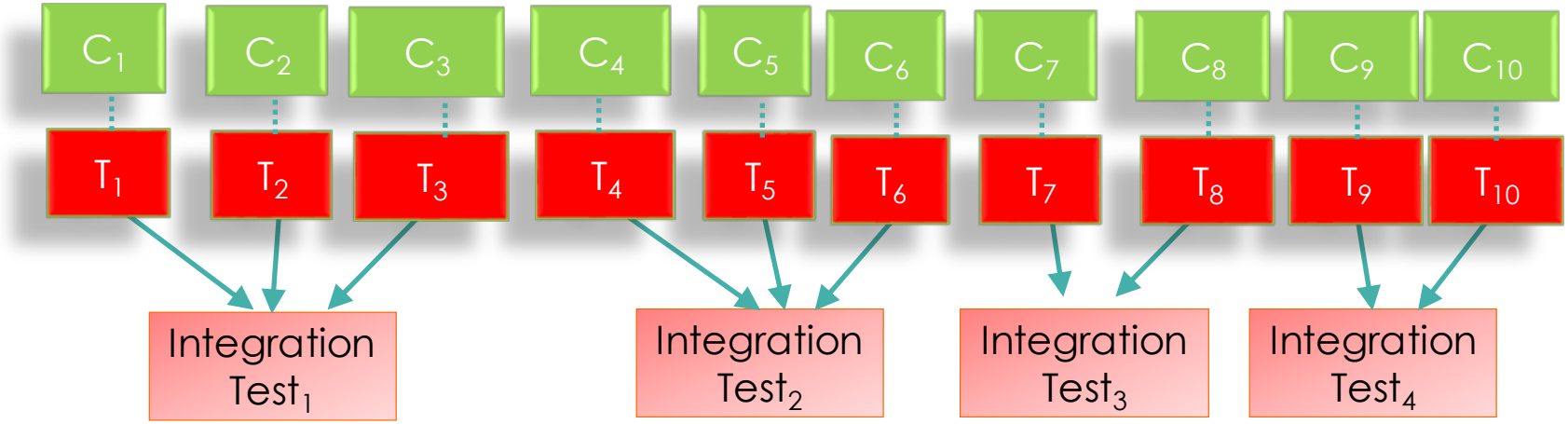
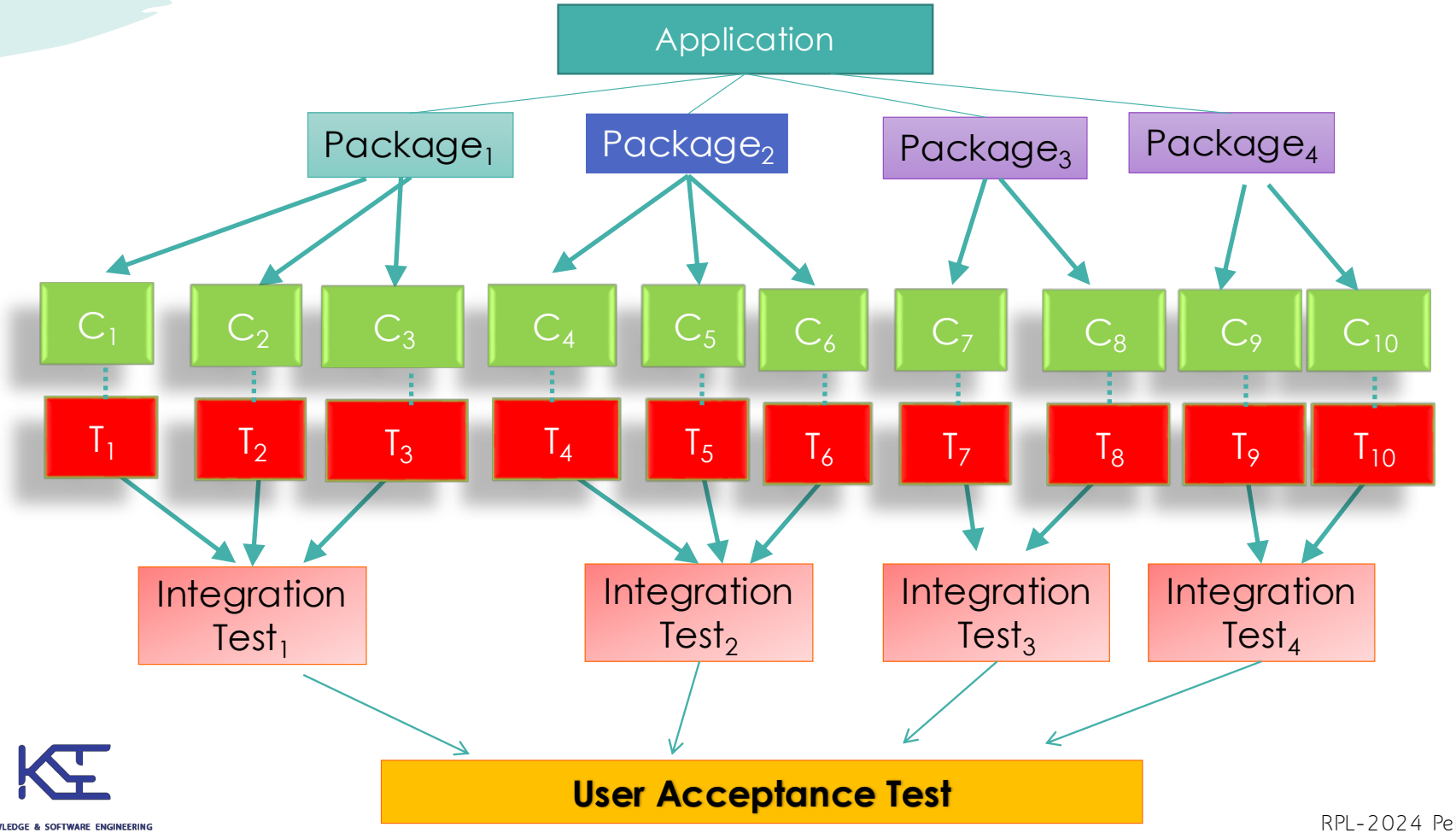
Integration Testing…



User Acceptance Testing…



Pra Tugas – Pembentukan Kelompok

- Buatlah kelompok dengan anggota 5 mahasiswa
  - Ketua kelas bertanggung jawab memberikan nomor kelompok
  - Ketua kelas akan mengirim nomor kelompok dan nama anggotanya via MS Teams kepada dosen kelas.
- Pilihlah topik untuk tugas pengembangan aplikasi di kuliah Rekayasa Perangkat Lunak ini

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